NC STATE UNIVERSITY

University Catalog 2026-2027

Analytics



The Master of Science in Analytics (MSA) is uniquely designed to equip students for the task of deriving and effectively communicating actionable insights from a vast quantity and variety of data. It is an intensive 10-month degree with a strong practical orientation

focused on the tools and methods used by data scientists. It is a fully integrated course of study taught exclusively to MSA students and designed to produce well-rounded professionals. Student teams tackle genuine problems with data provided by industry and government sponsors.

Master's Degree Requirements

Students complete 30 credit hours of defined coursework in a period of ten months beginning in Summer Session II and ending the following Spring semester. The integrated curriculum is designed to provide a focused education in the software tools, methods and applications of data analytics.

Other Relevant Information

Students must begin the degree program in the first semester (Summer Session II) and complete all 30 credit hours of the curriculum. The program is designed for full-time students only. Applications for admission are reviewed between September and April.

2026-2027 Program Schedule

Summer II 2026: AA 500 and AA 501

- Start date: June 29, 2026
- Census date: July 1, 2026
- End date: July 31, 2026
- Communication Training (required): August 3 - August 7, 2026

Fall 2026: AA 502 and AA 504

- Start date: August 17, 2026
- Census date: August 28, 2026
- End date: November 30, 2026

- Practicum project work, midpoint presentations, career and professional development activities (required): December 1-11, 2026

Spring 2027: AA 503 and AA 505

- Start date: January 11, 2027
- Census date: January 25, 2027
- End date: April 29, 2027
- Spring Commencement: May 8, 2027

More Information

Visit the Institute for Advanced Analytics website for details.

Admission Requirements

Admission to the MSA program is highly competitive. The best-qualified applicants will be accepted up to the limited number of seats available for students each year. The admissions committee evaluates candidates on criteria such as:

- overall academic record and grade point average;
- academic performance in analytical/quantitative subjects;
- relevant employment experience and potential to succeed in the profession; and
- leadership potential, integrity, and other personal character traits.

The Institute welcomes applications from highly motivated individuals of exceptional talent regardless of undergraduate major. Applicants without prior coursework in statistics and/or experience with computer programming would need to complete a set of prerequisite courses before qualifying as a candidate for admission.

Applicant Information

- **Delivery Method:** On Campus
- **Entrance Exam:** None
- **Interview Required:** Yes

Application Deadlines

- Summer II: Please visit the MSA program website for application timelines for US citizens/permanent residents and international applicants.

Degrees

- Analytics (MS)

Full Professors

William M. Rand

Practice/Research/Teaching Professors

Hunter S. Glanz

Susan Jeanne Simmons

Christopher West

Andrea Villanes Arellano

Sarah Egan Warren

Courses

AA 500 Analytics Tools and Techniques (3 credit hours)
This course equips the student with basic and advanced computer programming skills needed to use industry-standard analytics tools for data analysis, including but not limited to: data access and management, data cleaning, data mining, text mining, geospatial analytics, forecasting, and optimization. Restricted to AA majors.

Corequisite: AA 501

Typically offered in Summer only

AA 501 Analytics Foundations (3 credit hours)
This course equips the student with basic knowledge of statistics required for further study in analytics. Topics include, but are not limited to: Exploratory Data Analysis, Linear Regression, Multiple Linear Regression, Regression Diagnostics, Logistic Regression, ANOVA, Cluster Analysis, Analysis of Tables, and Survey Data Analysis. Restricted to AA major.

Corequisite: AA 500

Typically offered in Summer only

AA 502 Analytics Methods and Applications I (6 credit hours)
This course equips the students with the methods and applications of advanced analytics. Topics include, but are not limited to: Time Series and Forecasting, Geospatial Data Analytics, Linear Algebra, Data Mining, Survival Data Analysis and Logistic Regression Models. Restricted to AA major.

Prerequisite: AA 500 and AA 501; Corequisite: AA 504

Typically offered in Fall only

AA 503 Analytics Methods and Applications II (6 credit hours)
This course equips the student with the methods and applications of advanced analytics. Topics include, but are not limited to: Advanced Data Mining, Text Mining, Financial Analytics, Risk Analytics, Marketing Science and Customer Analytics, Linear and Non-Linear Programming. Restricted to AA major.

Prerequisite: AA 502; Corequisite: AA 505

Typically offered in Spring only

AA 504 Analytics Practicum I (6 credit hours)
This course equips the student with the knowledge and skills needed to conduct and present large-scale studies based on advanced analytics. Student teams conduct

analysis using large amounts of real-world data. Restricted to AA major.

Prerequisite: AA 500 and AA 501; Corequisite: AA 502

Typically offered in Fall only

AA 505 Analytics Practicum II (6 credit hours)

A continuation of AA 504, this course equips the student with the knowledge and skills needed to conduct and present large-scale studies based on advanced analytics. Student team conduct analysis using large amounts of real-world data. Restricted to AA majors.

Prerequisite: AA 504; Corequisite: AA 503

Typically offered in Spring only

AA 591 Special Topics in Advanced Analytics (1-6 credit hours)

Special Topics in Advanced Analytics

AA 691 Special Topics in Advanced Analytics (1-6 credit hours)

Special Topics in Advanced Analytics

NC STATE Academic Catalog 2026-2027

Department of Registration and Records
2831 Thurman Drive
Harris Hall, Room 1000
Campus Box 7313
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